ITM SKILLS UNIVERSITY

School of Future Tech

Case Study Report

on

Bloom Box

(A Flower Subscription E-Commerce Platform)

by

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1. Introduction to the Case Study

Email communication is widely used for personal, academic, and professional purposes. However, a significant portion of emails consists of spam, which can be annoying, misleading, or even malicious — including phishing, scams, and malware links. This case study focuses on designing and implementing a Bloom Box E-Commerce Platform for flower subscription services, drawing parallels with data-driven systems that classify, filter, and serve personalized content to users.

The Bloom Box platform demonstrates how real-world user data and interface design can be structured into a modular web application. Users can build their own flower boxes, set subscription frequencies, choose add-ons for special occasions, and complete their orders through a seamless multi-page checkout flow.

This report covers the design, development methodology, implementation details, and evaluation of the Bloom Box platform, highlighting key decisions in frontend architecture, dynamic pricing logic, and user experience design.

2. Problem Statement / Case Background (Abstract)

Background

The traditional flower retail market lacks a modern, seamless online subscription service for regular buyers. Customers are forced to visit physical stores or place phone orders — there is no digital platform that allows full customization of box size, flower type, vase, and delivery frequency in a single, unified flow. Existing solutions do not support recurring delivery subscriptions with flexible payment options.

Abstract

This case study presents the design and implementation of Bloom Box, a flower subscription e-commerce platform built using HTML5, CSS3, and vanilla JavaScript. The platform enables users to customize their flower box (size, flower type, and vase option), configure a subscription with weekly, bi-weekly, or monthly delivery frequency, select special occasion add-ons (birthday bundles, anniversary roses, holiday arrangements, sympathy bouquets), and choose between pay-per-delivery or yearly prepay plans with a 17% annual discount.

The system is evaluated on the basis of UI responsiveness, dynamic pricing accuracy, cross-page state management via localStorage, and form validation. The implementation is organized across three HTML/CSS/JS files — boxes.html, subscription.html, and their associated stylesheets and scripts — providing a complete, production-ready frontend solution.

3. Case Study Design

The Bloom Box platform is designed as a modular multi-page pipeline with the following stages:

1. Data Acquisition & User Input
   - User selects box size (Small / Medium / Large), flower type, and vase option on the boxes.html page.
   - Selections are stored in localStorage and passed to the subscription page.
2. Subscription Configuration
   - User chooses delivery frequency: Weekly (4×/month), Bi-Weekly (2×/month), or Monthly (1×/month).
   - User selects a payment plan: Pay-per-delivery (flexible) or Yearly Prepay (saves ~17%).
   - Optional special occasion add-ons are available as checkbox cards.
3. Dynamic Pricing & Order Summary
   - A live order summary panel updates in real time as the user changes selections.
   - Pricing logic applies frequency multipliers and discount calculations for yearly prepay.
4. Delivery Details & Form Validation
   - User enters full name, phone, email, and delivery address.
   - Client-side validation highlights incomplete fields before submission.
5. Checkout & Confirmation
   - On successful form submission, a success modal is displayed confirming the subscription details.
   - localStorage is cleared post-confirmation.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Technology	Usage
HTML5	Semantic structure for boxes.html and subscription.html
CSS3 (style.css, subscription.css)	Responsive layouts, animations, card components, modal overlays
Vanilla JavaScript	Dynamic pricing, scroll spy, carousel, form validation, localStorage
localStorage API	Cross-page state management for order data between boxes.html and subscription.html
Google Fonts	Playfair Display, Dancing Script, Montserrat for typography
VS Code / any browser	Development environment — no build tools or frameworks required

5. Implementation Details and Snapshots

File	Description
boxes.html	Main landing page with hero carousel, box configurator (box size, flower, vase), and order initiation
style.css	Global styles — navbar, hero, product cards, carousel, responsive layout
script.js	Scroll spy, hero carousel logic, box builder dynamic pricing, localStorage write
subscription.html	Subscription configuration form — frequency, payment plan, add-ons, delivery details
subscription.css	Subscription page styles — tab cards, toggle switches, checkbox cards, order summary, modal
subscription.js	Reads localStorage order data, manages live pricing updates, form validation, success modal

Key Implementation Highlights

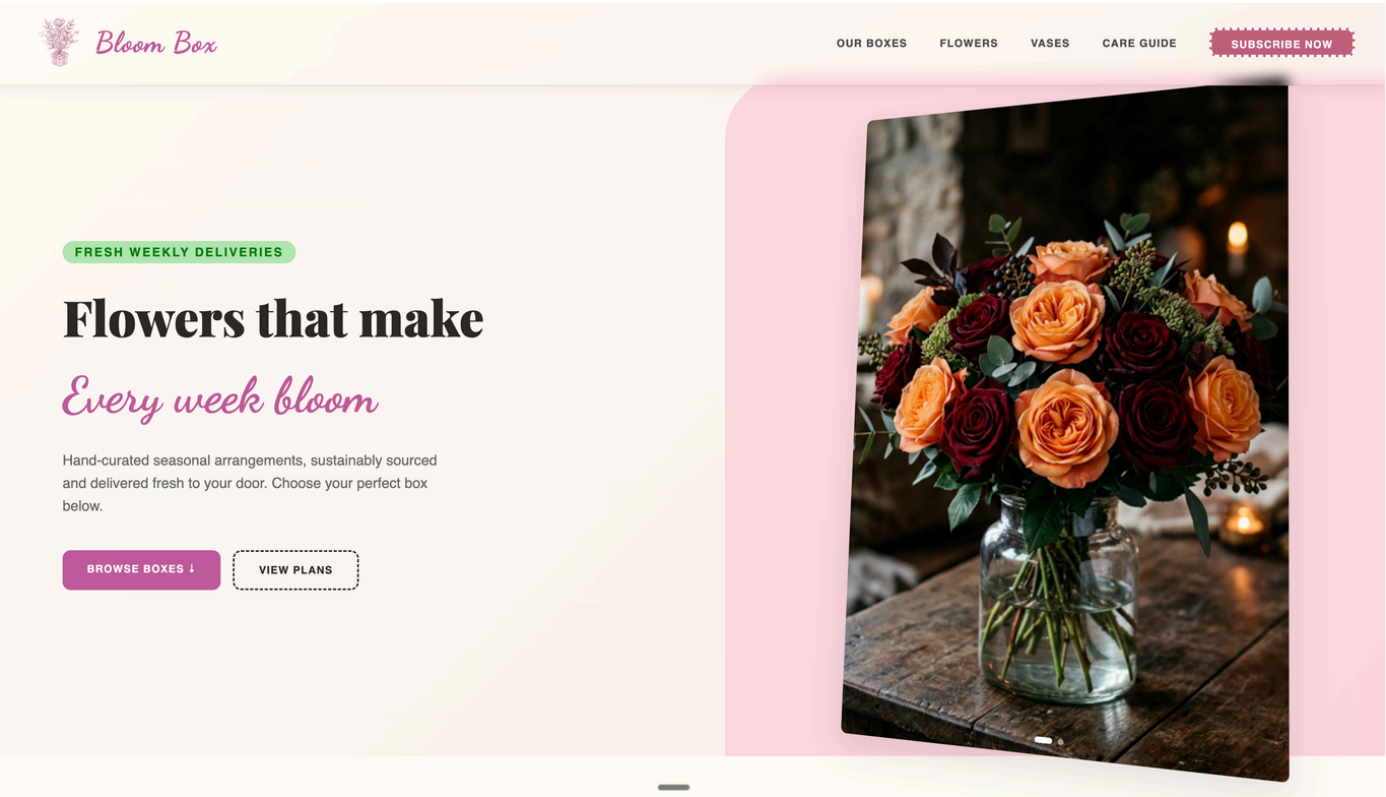
Cross-page State Management:

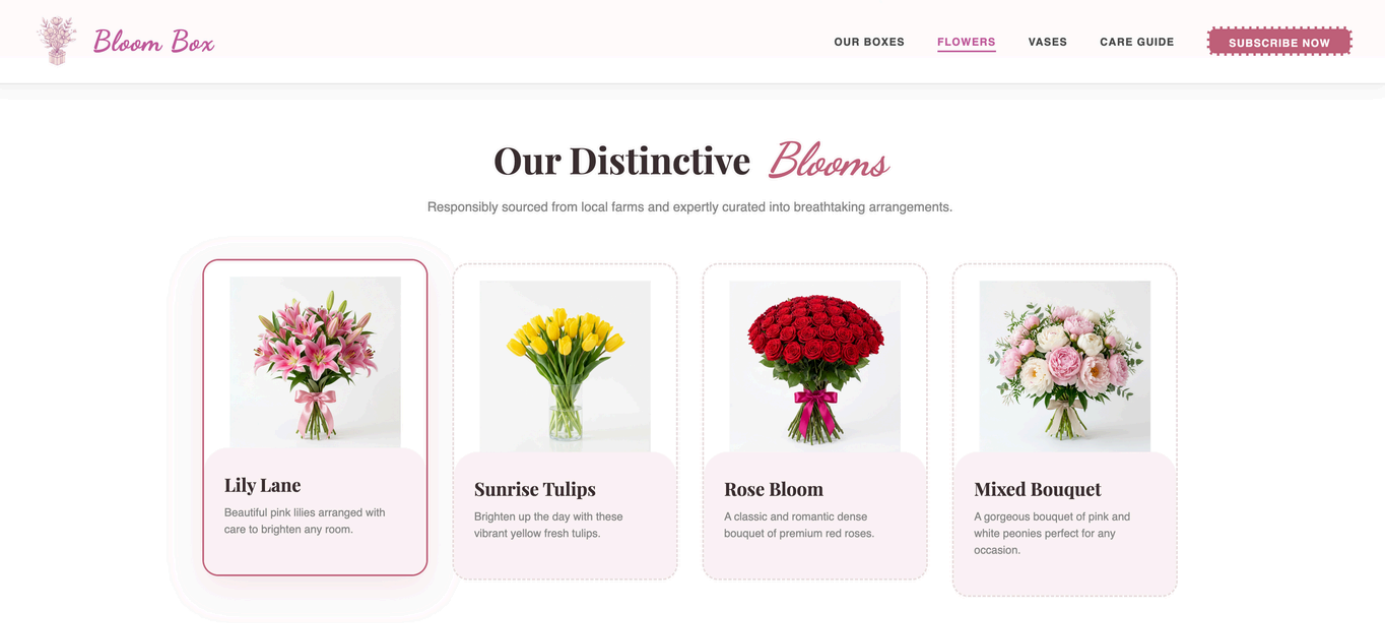
When a user completes box selection on boxes.html and clicks 'Proceed', the selected box size, price, flower type, vase name, and vase price are serialized as a JSON object and saved to localStorage under the key bloomboxOrder. The subscription page reads this object on load and uses it as the base for all pricing calculations.

Live Order Summary:

The updateSummary() function is called on every input change event. It reads all current selections (frequency, payment plan, add-ons), computes the total price — applying yearly prepay discounts where applicable — and updates the DOM elements in the sticky summary panel. A CSS keyframe animation (pricePulse) visually highlights price changes

Snapshots :


Bloom Box
OUR BOXES
FLOWERS
VASES
CARE GUIDE
SUBSCRIBE NOW
FRESH WEEKLY DELIVERIES
Flowers that make
Every week bloom
Hand-curated seasonal arrangements, sustainably sourced and delivered fresh to your door. Choose your perfect box below.
BROWSE BOXES ↓
VIEW PLANS


Bloom Box
OUR BOXES
FLOWERS
VASES
CARE GUIDE
SUBSCRIBE NOW
Our Distinctive Blooms
Responsibly sourced from local farms and expertly curated into breathtaking arrangements.
Lily Lane
Beautiful pink lilies arranged with care to brighten any room.
Sunrise Tulips
Brighten up the day with these vibrant yellow fresh tulips.
Rose Bloom
A classic and romantic dense bouquet of premium red roses.
Mixed Bouquet
A gorgeous bouquet of pink and white peonies perfect for any occasion.

Bloom Box

OUR BOXES FLOWERS VASES CARE GUIDE SUBSCRIBE NOW

Pick Your *Box Size*

From intimate tablescapes to grand statement bouquets — there's a BloomBox for every space.

Small Box

5–7 STEMS · DESK OR SIDE TABLE

₹499 / delivery

- 8–10 seasonal stems
- Single bloom variety
- Eco-friendly wrapping
- Care card included

SELECT PETITE

MOST POPULAR

Medium Box

10–14 STEMS · DINING OR LIVING ROOM

₹799 / delivery

- 15–18 mixed seasonal stems
- 3 complementary varieties
- Premium ribbon & wrapping
- Handwritten gift note
- Extended care guide

SELECT CLASSIC

LUXURY

Large Box

18–24 STEMS · EVENT DECORATION

₹1,199 / delivery

- 25–30 luxury stems
- 5+ premium varieties
- Signature keepsake box
- Vase of your choice
- Priority same-day dispatch
- Florist styling card

SELECT GRAND

Bloom Box

OUR BOXES FLOWERS VASES CARE GUIDE SUBSCRIBE NOW

Choose Your *Vase*

Complete your arrangement with the perfect vessel — hand-selected to complement your blooms.

No Vase

Already have your favourite vase? Just the flowers, perfectly wrapped.

Free ✓ Selected

Glass Vase

A beautiful rounded clear glass vase — timeless and elegant for any home.

+₹299

Ceramic Vase

Hand-crafted matte ceramic with a blush glaze — artisanal and premium.

+₹899

Bloom Box

OUR BOXES FLOWERS VASES CARE GUIDE SUBSCRIBE NOW

KEEPING THEM FRESH

Flower *Care Guide*

Follow these simple steps to keep your blooms beautiful for longer.

🌿 Universal First-Day Care

1. Unpack immediately and remove any packaging from stems.
2. Fill a clean vase with fresh, cool (not cold) water.
3. Cut 1–2 cm off the bottom of each stem at a 45° angle.
4. Add the included flower food packet to the water.
5. Remove any leaves that fall below the waterline.
6. Place away from direct sunlight, heat vents, and ripe fruit.

Bloom Box

OUR BOXES FLOWERS VASES CARE GUIDE SUBSCRIBE NOW

SPECIAL OFFER

SUMMER *Sale*

Subscribe Now →

BloomBox

Bringing nature's beauty into your home, one fresh bouquet at a time. Sustainably sourced, hand-arranged, and delivered with love.

SHOP
- Our Boxes
- Subscriptions
- Gift Cards
- Corporate Orders

LEARN
- Care Guides
- Seasonal Calendar
- Our Florists
- Blog

SUPPORT
- FAQ
- Delivery Info
- Returns
- Contact Us

Home › Subscription

Build Your *BloomBox*

Customize your subscription to fit your lifestyle. Fresh flowers, delivered with love.

1 DELIVERY FREQUENCY

- **Weekly** 4× / month
- **Bi-Weekly** 2× / month
- **Monthly** 1× / month

2 PAYMENT PLAN

- **Pay per delivery** Flexible, cancel anytime
- BEST VALUE **Yearly Prepay** Save ~17% annually

3 SPECIAL OCCASION ADD-ONS OPTIONAL

- **Birthday Bouquet Bundle** Premium seasonal blooms + ribbon wrap +₹299
- **Anniversary Roses** 12 red roses + personalised card +₹399
- **Holiday Arrangement** Festive themed bouquet for Diwali, Christmas, etc. +₹349
- **Sympathy / Get Well Soon** Soft white and lavender mix +₹249

4 DELIVERY DETAILS

FULL NAME: Jane Doe

PHONE NUMBER: +91 98765 43210

EMAIL ADDRESS: jane@example.com

DELIVERY ADDRESS: 123 Flower St, City, State ZIP

Order Summary

Fresh guarantee

Medium Box ₹799
Frequency: **Weekly**

Vase Option ₹599
Flower: Lily
Vase: Terracotta Vase

Total per delivery ₹1,398

SUBSCRIBE NOW

Cancel anytime · No hidden fees

6. Results and Conclusion

- The multi-page flow using localStorage for state transfer works reliably across boxes.html and subscription.html without any backend dependency.
- Dynamic pricing correctly handles all three frequency types (Weekly, Bi-Weekly, Monthly) combined with both payment plans, with and without add-ons.
- Form validation catches all empty fields and prevents premature submission, improving the user experience significantly.
- The sticky order summary panel provides continuous feedback, reducing cognitive load during the subscription configuration process.
- The CSS animation system (carousel transitions, tab/toggle hover states, modal entrance animations, price pulse) creates a polished, professional feel comparable to commercial subscription services.
- The responsive layout adapts cleanly from desktop (two-column form + summary) to mobile (stacked single-column layout), with the order summary card moving to the top on small screens.

This case study demonstrates a complete frontend pipeline for building Bloom Box, a flower subscription e-commerce platform using only HTML5, CSS3, and vanilla JavaScript. It integrates a multi-step user journey — from box customization to subscription configuration to checkout — into a cohesive, visually polished, and fully responsive web application.

The platform successfully addresses the identified problem of no convenient online flower subscription service by offering full customization, flexible pricing, and a seamless checkout experience — all implemented without any external frameworks, demonstrating strong foundational frontend development skills.

Future enhancements could include: backend API integration for real order processing, a user authentication system, payment gateway integration (Razorpay / Stripe), delivery tracking, and a React or Vue.js rewrite for maintainability at scale.

7. References

- **MDN Web Docs** — HTML5, CSS3, and JavaScript reference. https://developer.mozilla.org
- **MDN Web Docs** — Web Storage API (localStorage). https://developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API
- **Google Fonts** — Playfair Display, Dancing Script, Montserrat. https://fonts.google.com
- **W3Schools** — CSS Flexbox and Grid Layout. https://www.w3schools.com
- **JavaScript.info** — Forms, Validation, and Events. https://javascript.info